

TOSHIBA TRANSISTOR SILICON NPN EPITAXIAL TYPE

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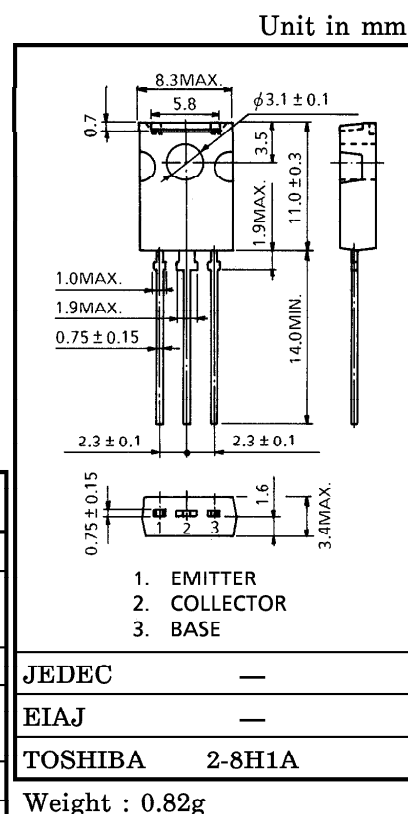
STROBE FLASH APPLICATIONS.

MEDIUM POWER AMPLIFIER APPLICATIONS.

- High DC Current Gain
: $h_{FE}(1) = 800 \sim 3200$ ($V_{CE} = 2V$, $I_C = 0.5A$)
: $h_{FE}(2) = 250$ (Min.) ($V_{CE} = 2V$, $I_C = 4A$)
- Low Saturation Voltage
: $V_{CE(sat)} = 0.5V$ (Max.) ($I_C = 4A$, $I_B = 40mA$)
- High Collector Power Dissipation
: $P_C = 10W$ ($T_c = 25^\circ C$), $P_C = 1.5W$ ($T_a = 25^\circ C$)

MAXIMUM RATINGS ($T_a = 25^\circ C$)

CHARACTERISTIC		SYMBOL	RATING	UNIT
Collector-Base Voltage		V_{CBO}	50	V
Collector-Emitter Voltage		V_{CES}	40	V
		V_{CEO}	20	
Emitter-Base Voltage		V_{EBO}	8	V
Collector Current	DC	I_C	5	A
	Pulse (Note 1)	I_{CP}	8	
Base Current		I_B	0.5	A
Collector Power Dissipation	$T_a = 25^\circ C$	P_C	1.5	W
	$T_c = 25^\circ C$		10	
Junction Temperature		T_j	150	$^\circ C$
Storage Temperature Range		T_{stg}	$-55 \sim 150$	$^\circ C$



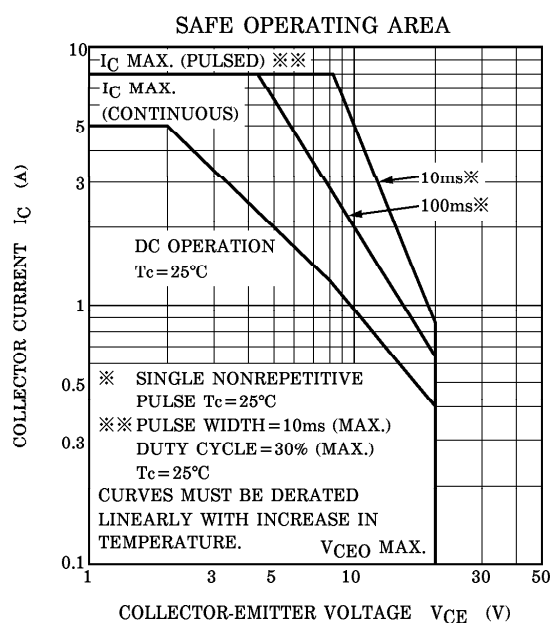
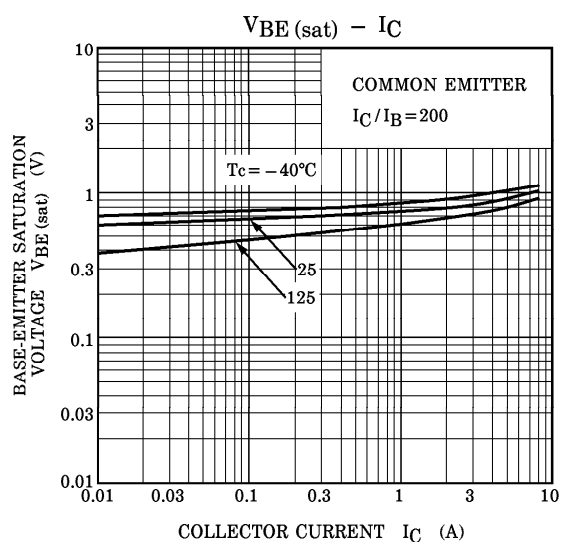
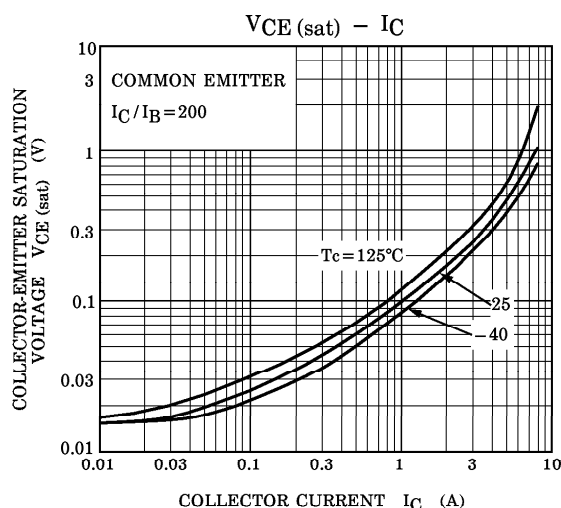
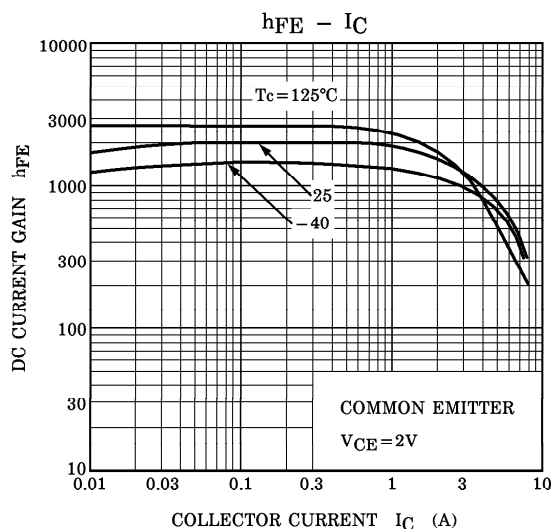
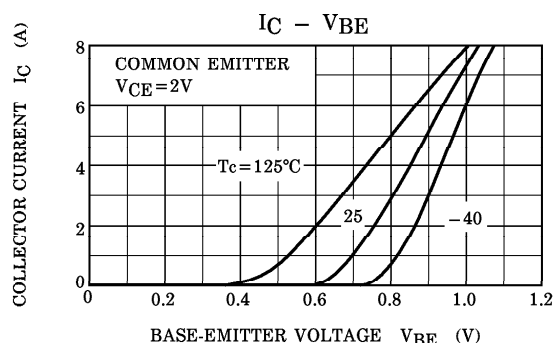
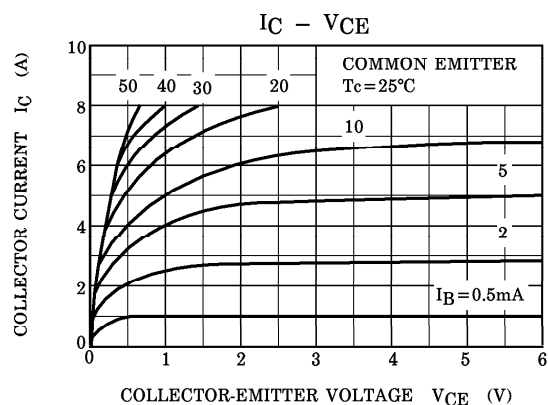
Note 1 : Pulse Test : Pulse Width=10ms (Max.) Duty Cycle=30% (Max.)

ELECTRICAL CHARACTERISTICS ($T_a = 25^\circ C$)

CHARACTERISTIC	SYMBOL	TEST CONDITION	MIN.	TYP.	MAX.	UNIT
Collector Cut-off Current	I_{CBO}	$V_{CB} = 50V$, $I_E = 0$	—	—	100	nA
Emitter Cut-off Current	I_{EBO}	$V_{EB} = 8V$, $I_C = 0$	—	—	100	nA
Collector-Emitter Breakdown Voltage	$V_{(BR) CEO}$	$I_C = 10mA$, $I_B = 0$	20	—	—	V
DC Current Gain	$h_{FE}(1)$	$V_{CE} = 2V$, $I_C = 0.5A$	800	—	3200	
	$h_{FE}(2)$	$V_{CE} = 2V$, $I_C = 4A$	250	—	—	
Collector-Emitter Saturation Voltage	$V_{CE(sat)}$	$I_C = 4A$, $I_B = 40mA$	—	—	0.5	V
Base-Emitter Voltage	V_{BE}	$V_{CE} = 2V$, $I_C = 4A$	—	—	1.2	V
Transition Frequency	f_T	$V_{CE} = 2V$, $I_C = 0.5A$	—	150	—	MHz
Collector Output Capacitance	C_{ob}	$V_{CB} = 10V$, $I_E = 0$, $f = 1MHz$	—	45	—	pF

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